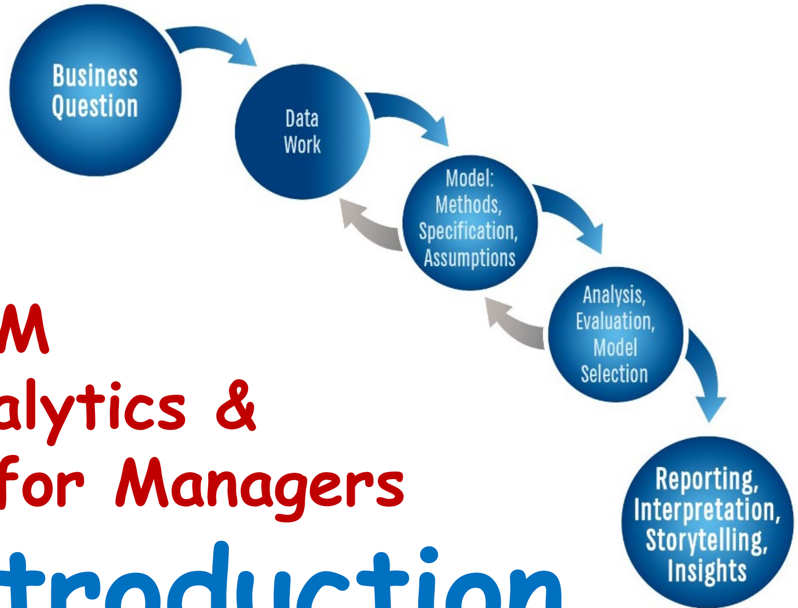

PAML4M
Predictive Analytics & Machine Learning for Managers

1. Course Introduction

Scripts:

PAML4M_1R_Packages.Qmd (Github)
PAML4M_1R_R_Intro.Qmd (Github)
ITEC_Quarto.Qmd (Canvas)

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Last updated 8/1/2026

KOGOD SCHOOL *of* BUSINESS
AMERICAN UNIVERSITY • WASHINGTON, DC

A Few Popular Quotes

"In God we trust. All others must bring data"
W. Edwards Deming, engineer and statistician

"If statistics are boring, you've got the wrong numbers"
Peter Drucker, management author

"Torture the data, and it will confess to anything"
Ronald Coase, Economics, Nobel Prize

"He uses statistics as a drunken man uses lamp posts – for support rather than illumination"
Andrew Lang, Scottish poet and anthropologist, 1844-1912

Billy Beane (Oakland A's) incorporated ("**sabermetrics**") analytics and statistics, reaching record profits in baseball


Billy Beane

William Lamar Beane III (born March 29, 1962) is a former American professional baseball player and current front office executive. He is the executive vice president of baseball operations and minority owner of the Oakland Athletics of Major League Baseball

A first-round pick in the MLB draft by the Mets, Beane failed to meet the expectations of scouts, who projected him as a star. In his front-office career, Beane has applied statistical analysis (known as sabermetrics) to baseball, which has led teams to reconsider how they evaluate players. He is the subject of Michael Lewis's 2003 book on baseball economics, *Moneyball*, which was made into a 2011 film starring Brad Pitt as Beane.

Bill James

In 2006, *Time* named him in the Time 100 as one of the most influential people in the world.[3] He is a Senior Advisor on Baseball Operations for the Boston Red Sox. In 2010, Bill James was inducted into the Irish American Baseball Hall of Fame.[4]

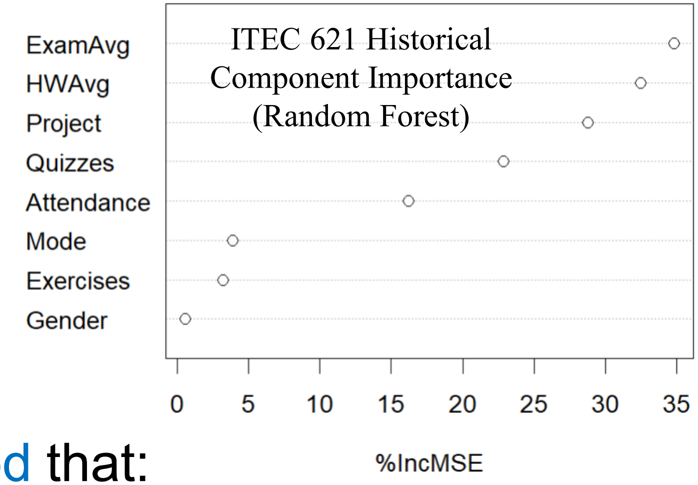
His approach, which he termed sabermetrics in reference to the Society for American Baseball Research (SABR),[1] scientifically analyzes and studies baseball, often through the use of statistical data, in an attempt to determine why teams win and lose.

Key Learning Objectives

- We will cover all aspects of the analytics cycle, but with an emphasis on predictive modeling and interpretation
- This is NOT a Statistics class, but it requires an understanding of Statistics and Machine Learning
- This is NOT a Programming class, but it requires the ability to use analytics tools, like

The main learning objectives in this course are to learn how to identify, select, justify, and employ the most appropriate model to answer a business analytics question and provide meaningful managerial interpretations of your results

Key Knowledge & Skills



To succeed as an analytics professional:

1. **Frame** a compelling business question
2. **Gather** and pre-process, transform and prepare the data for analysis.
3. **Select** and apply the best modeling method that:
   - ✓ Meets your analysis goals (i.e., interpretation, prediction, inference)
4. **Identify** the best model specification (e.g., variable selection, feature engineering, etc.) for your modeling approach
   - ✓ Of all viable models and specifications, select the ones that yield the best predictive accuracy (i.e., machine learning/cross-validation)
5. **Interpret** and **explain** your results to your target audience

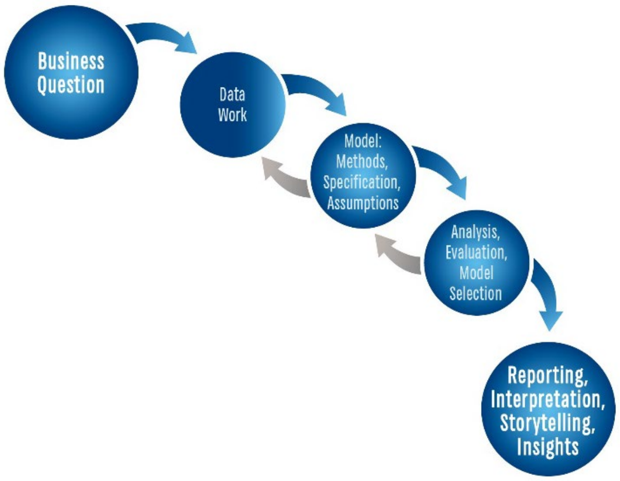
Key → You cannot learn everything about analytics in one course.
The goal in this class is to learn how to continue learning on your own !!

Analytics

"It is the scientific process of transforming data into insight for making better decisions" (INFORMS)

Analytics Question → Analysis → Insights → Storytelling

In analytics, the starting point is a specific business question you want to answer or hypotheses you want to test and end with a **compelling answer** to that question.



Machine Learning

Is a method of data analysis that uses algorithms that learn from data iteratively allowing computers to find hidden insights without being explicitly programmed (SAS: http://www.sas.com/en_us/insights/analytics/machine-learning.html)

- **Unsupervised** learning → data exploration without specific goals (closely associated with descriptive analytics and data mining, e.g., clustering, correlation analysis, histograms)
- **Supervised** learning → data analysis with specific goals in mind → training predictive models with existing data → and constantly testing their predictive accuracy with existing data

First, develop familiarity with the data

Descriptive Analytics

PizzaID	brand	mois	prot	fat	ash	sodium	carb	cal
14001	D	47.17	22.29	21.3	4.08	0.74	5.16	302
14002	D	49.16	27.99	17.49	3.29	0.39	2.07	278
14003	A	30.49	21.28	41.65	4.82	1.64	1.76	467
14004	B	52.68	14.38	25.72	3.26	0.93	3.96	305
14005	H	33.05	7.34	15.78	1.34	0.42	42.49	341
14006	H	35.55	7.32	16.4	1.76	0.36	38.97	333
14007	G	28.68	8.3	16.07	1.41	0.45	45.54	360

e.g., descriptive statistics, frequency distributions, correlation, ANOVA, clustering analysis, etc.

Then, develop predictive models

Predictors → ***Outcome***

PizzaID	brand	mois	prot	fat	ash	sodium	carb
14001	D	47.17	22.29	21.3	4.08	0.74	5.16
14002	D	49.16	27.99	17.49	3.29	0.39	2.07
14003	A	30.49	21.28	41.65	4.82	1.64	1.76
14004	B	52.68	14.38	25.72	3.26	0.93	3.96
14005	H	33.05	7.34	15.78	1.34	0.42	42.49
14006	H	35.55	7.32	16.4	1.76	0.36	38.97
14007	G	28.68	8.3	16.07	1.41	0.45	45.54

cal
302
278
467
305
341
333
360

e.g., Ordinary Least Squares (OLS) regression,
Logistic regression, etc.

Then, use machine learning:

Split, Train, Test, Re-Sample

Training Subset: Train Model → ***Outcome***

PizzaID	brand	mois	prot	fat	ash	sodium	carb	cal
14001	D	47.17	22.29	21.3	4.08	0.74	5.16	302
14002	D	49.16	27.99	17.49	3.29	0.39	2.07	278
14003	A	30.49	21.28	41.65	4.82	1.64	1.76	467
14004	B	52.68	14.38	25.72	3.26	0.93	3.96	305

Testing Subset: Test Model's Accuracy → ***Outcome***

								Actual	Predicted	Error
14005	H	33.05	7.34	15.78	1.34	0.42	42.49	341	290	51
14006	H	35.55	7.32	16.4	1.76	0.36	38.97	333	362	-29
14007	G	28.68	8.3	16.07	1.41	0.45	45.54	360	371	-11

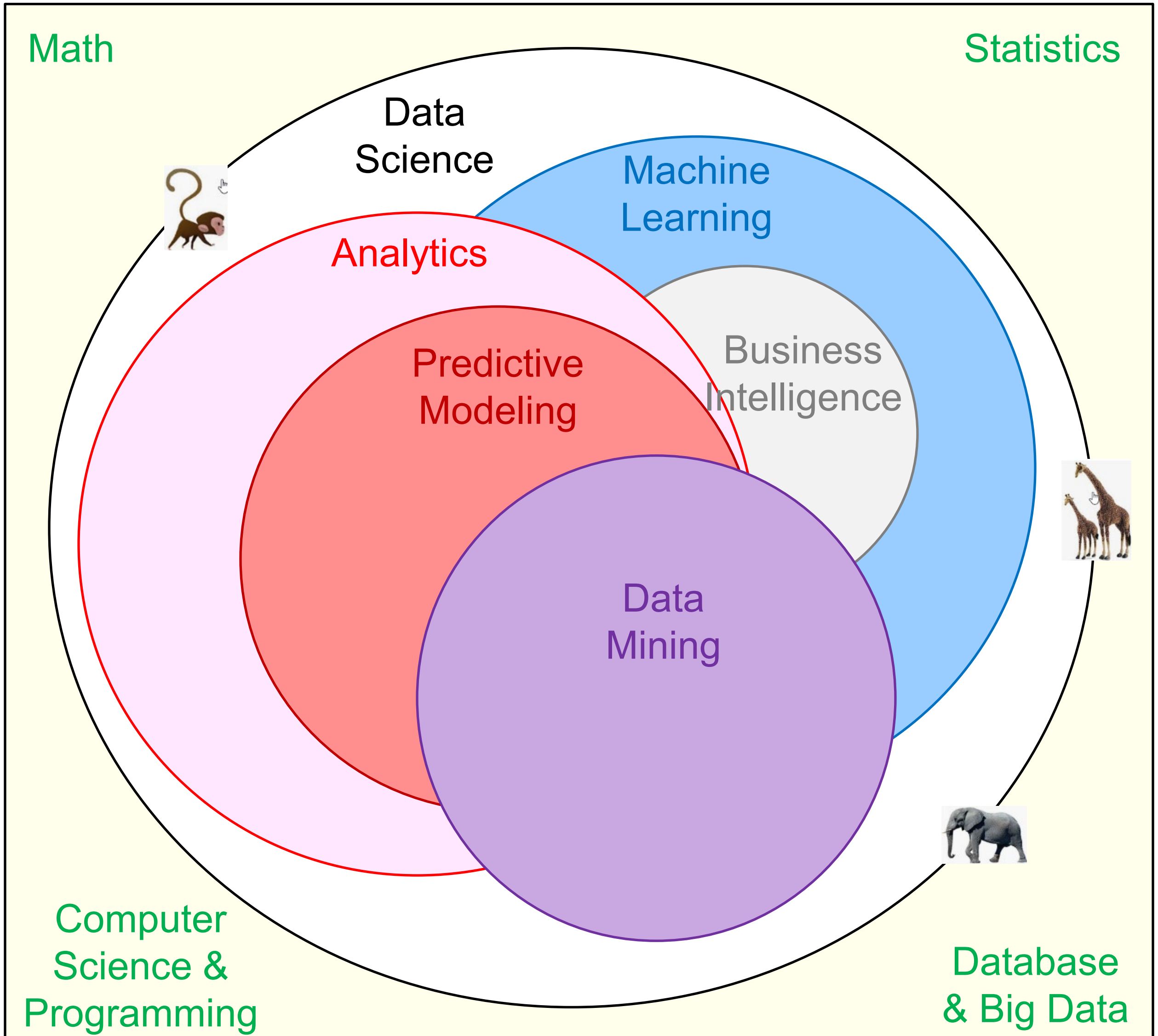
*e.g., **Cross-Validation** (CV) testing, tuning model parameters, deep learning models, Logistic regression, etc.*

Data Science

"Is a set of fundamental principles that guide the extraction of knowledge from data."
(Provost & Fawcett, Data Science for Business)

- A data scientist has deep knowledge on analytics, all aspects of data, the discipline or functional domain in which analysis is conducted (marketing, healthcare, social media, etc.), and the underlying core disciplines (e.g., statistics, mathematics, database, software programming, etc.)

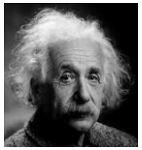
Analytics Zoo



Syllabus Overview

(See Canvas)

Predictive Analytics Overview



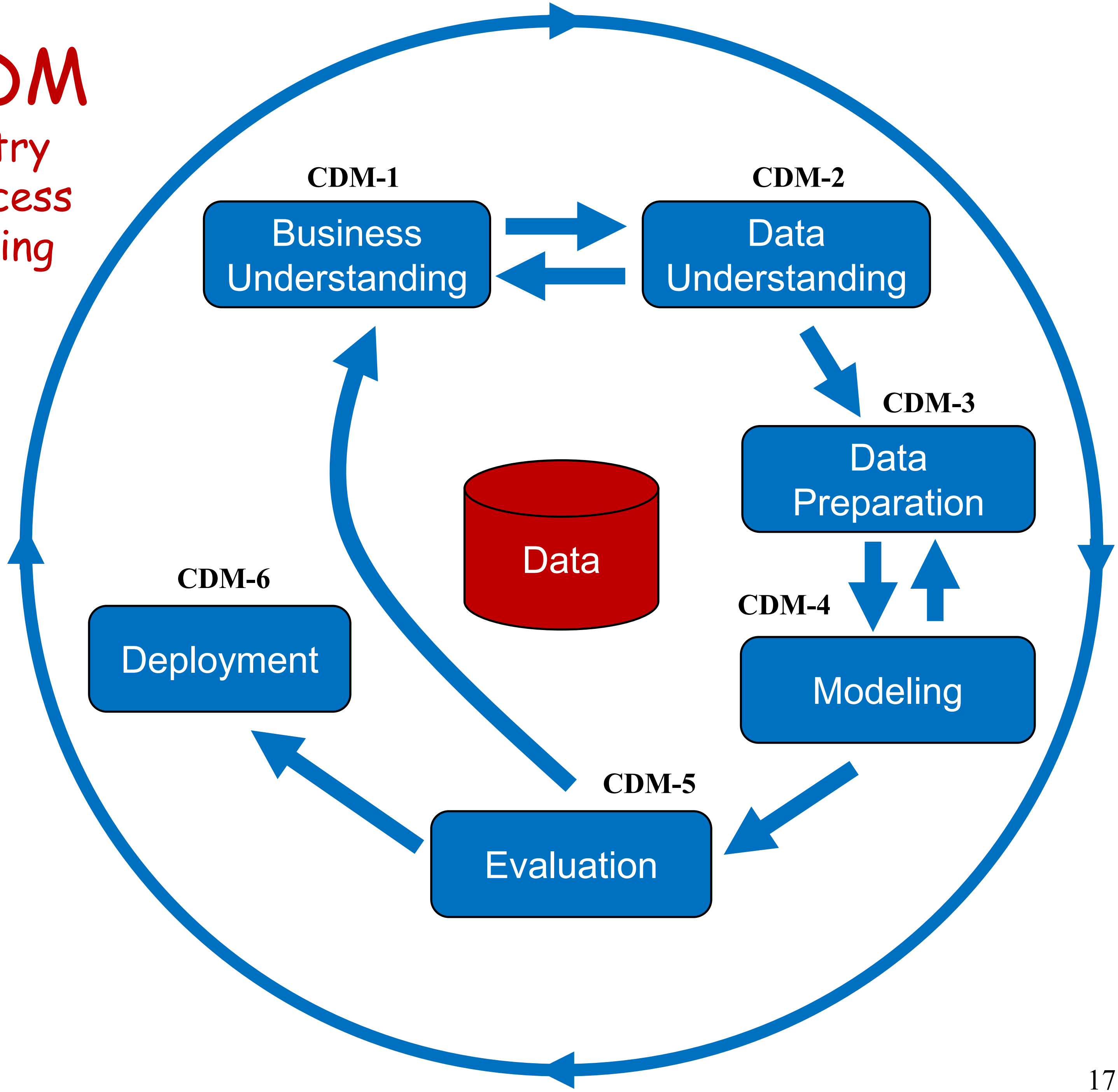
Food for Thought

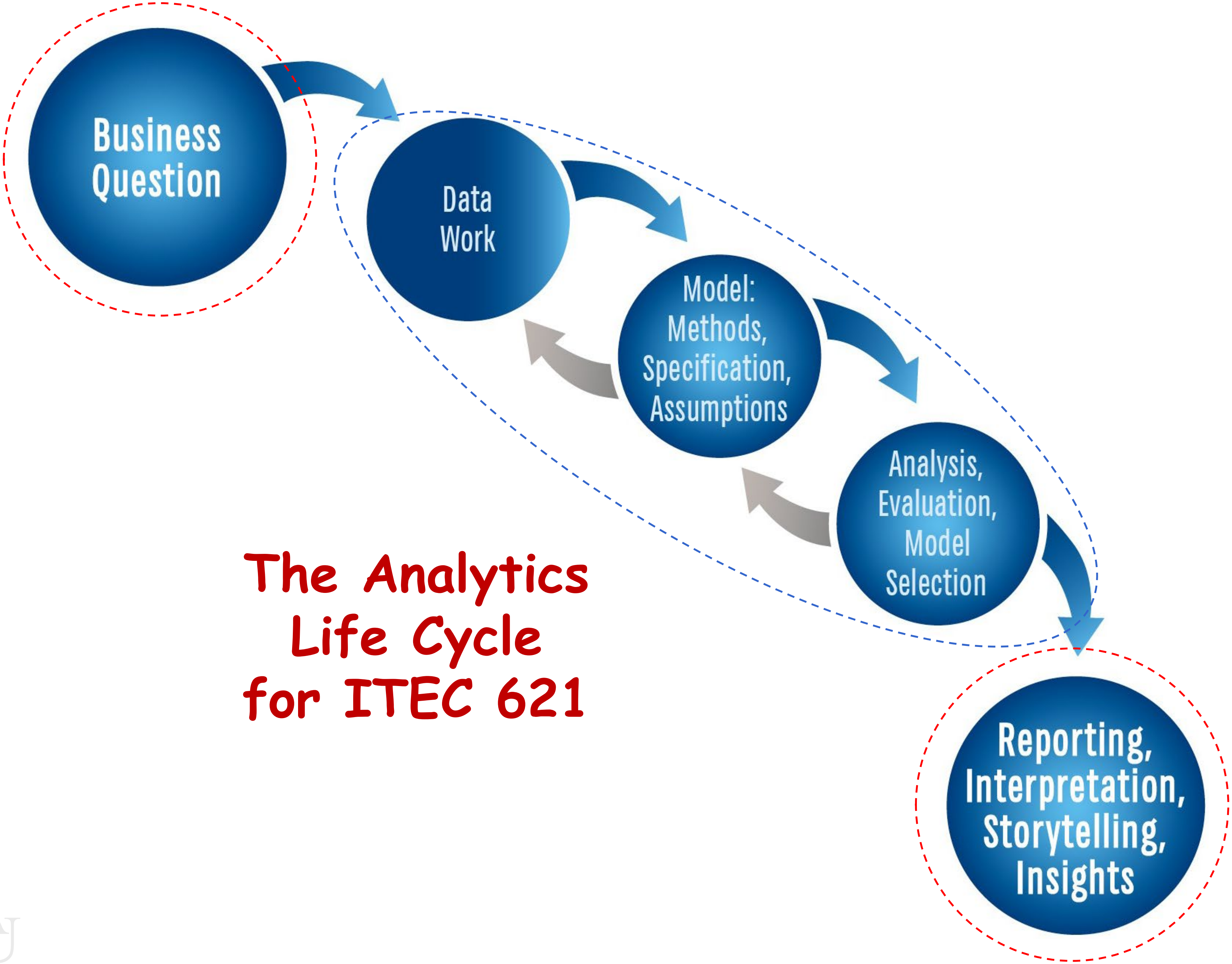
- Humans are living predictive analytics entities
- Politicians' tricks → single data point; single variable prediction
  - Does one case prove anything? Statistical power?
  - e.g., is there a gender pay gap?
  - ↑ $1 minimum wage → Impact on unemployment?
- What do doctors do? → diagnosis, treatment
- Why are you in the MS program? What are you predicting?
- Who in this class will be the first to become CEO?
- How much money will you make 5 years after graduation?
- Will you make more than $200K 5 years after graduation?
- What's the financial impact of providing free WiFi at a hotel?
- How to fight cybercrime, terrorism (Shining Path), fraud, etc.?
- Interesting cases: Target, Starbucks, Oakland A's

The Analytics Life Cycle

CRISP-DM

Cross-Industry Standard Process for Data Mining





The Analytics Life Cycle for ITEC 621

Step 1
Business & Analytics Question

The Business Question & Case

- **Business Question**:
  - ✓ Is the main project driver
  - ✓ Articulated in general and managerial terms

Examples: How can the spread of an epidemic be controlled?
What factors lead to increased sales in a store?

- **Business Case**:
  - ✓ Rationale for the importance of the business question
  - ✓ Why is it important for the business?
  - ✓ What is the value proposition?

The Analytics Question

Translate the business question into one or more analyzable questions. More specifically, define:

Sales ~
Advertising +
StoreSize +
SalesForce +
etc.

1. What is the outcome variable and its type?
    - **Quantitative** – e.g., salaries, sales, credit rating
    - **Classification** – e.g., loan default (yes/no), credit application (approved/declined) → binary vs. multiple categories
2. What are the key predictors of interest?
    - Focal predictors of interest – e.g., GPA, advertising expenditures, annual income, education level
    - Control variables – e.g., not of main interest, but may influence the outcome

Step 2
Data Work

Important Data Issues

- ETL → Extract (from multiple data sources), Transform (into structured formats) and Load (into analysis data environment)
- Data cleansing, missing values, data curation (i.e., organization and integration → often part of ETL)
- Pre-processing → necessary or optional data transformations for the analysis
- Data structures → vectors, matrices, data frames, lists, etc.
- Data types → numeric, binary, categorical, date, text, etc.
- Data notations → matrix notation, data models

Data Frames, Matrices & Vectors

Data Frame → *Pizzas*

PizzaID	brand	mois	prot	fat	ash	sodium	carb	cal
14001	D	47.17	22.29	21.3	4.08	0.74	5.16	302
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14007	G	28.68	8.3	16.07	1.41	0.45	45.54	360

- Columns are “vectors” (variables)
- Data across columns can be of different types
- Data within columns must be of same type

Matrix → *PizzaMat*

prot	fat	ash	sodium	carb	cal
22.29	21.3	4.08	0.74	5.16	302
27.99	17.49	3.29	0.39	2.07	278
21.28	41.65	4.82	1.64	1.76	467
14.38	25.72	3.26	0.93	3.96	305
7.34	15.78	1.34	0.42	42.49	341
7.32	16.4	1.76	0.36	38.97	333
8.3	16.07	1.41	0.45	45.54	360

- All data must be of same type
- Usually numeric

carb
5.16
2.07
1.76
3.96
42.49
38.97
45.54

Vector → *Pizzas$carb*

- Vectors can be extracted from data frames
- Vectors can be aggregated into data frames
- Data within a vector must be of same type

Matrix Notation

We will not use much matrix notation or algebra, but it helps to understand its basics and the textbook sometimes makes reference to it.

$$\overbrace{\underset{\mathbf{Y}\ (n \times 1)}{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \text{Etc,} \\ y_n \end{bmatrix}} \quad \underset{\mathbf{X}\ (n \times k)}{\begin{bmatrix} x_{11}\ x_{21}\ \ldots\ x_{k1} \\ x_{12}\ x_{22}\ \ldots x_{k2} \\ x_{13}\ x_{23}\ \ldots x_{k3} \\ \text{Etc,} \\ x_{1n}\ x_{2n}\ \ldots x_{kn} \end{bmatrix}}}^{\text{Data}} \rightarrow \overbrace{\underset{\boldsymbol{\beta}\ (1 \times k)}{\begin{bmatrix} \beta_1\ \beta_2\ \beta_3\ \text{Etc,}\ \beta_k \end{bmatrix}}}^{\text{Coefficients}} \rightarrow \overbrace{\underset{\hat{\mathbf{Y}}\ (n \times 1)}{\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \\ \text{Etc,} \\ \hat{y}_n \end{bmatrix}}}^{\text{Prediction}} + \overbrace{\underset{\boldsymbol{\varepsilon}\ (n \times 1)}{\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \text{Etc,} \\ \varepsilon_n \end{bmatrix}}}^{\text{Error}}$$

Predictors → Independent Variables (matrix)
(each variable is a vector)

Outcomes → Dependent Variable (vector)

Matrix algebra facilitates computation, e.g.:

Predictive Model: $Y = \hat{Y} + \varepsilon = \beta X + \varepsilon$

Sum of Squared Errors: $\varepsilon'\varepsilon$

Regression Coefficients: $\beta = (X' X)^{-1} X'Y$

More on Matrices: Covariance & Correlation

Covariance Matrix

$$\begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \ldots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} & \ldots & \sigma_{2n} \\ \sigma_{31} & \sigma_{32} & \sigma_3^2 & \ldots & \sigma_{3n} \\ \ldots & & & & \\ \sigma_{n1} & \sigma_{n2} & \sigma_{n3} & \ldots & \sigma_n^2 \end{bmatrix}$$

$\Sigma = \mathrm{Cov}(X)$

The scale of the data matters

$$\rho_{ij} = \sigma_{ij}/\sigma_i\sigma_j$$

Correlation Matrix

$$\begin{bmatrix} 1 & \rho_{12} & \rho_{13} & \ldots & \rho_{1n} \\ \rho_{21} & 1 & \rho_{23} & \ldots & \rho_{2n} \\ \rho_{31} & \rho_{32} & 1 & \ldots & \rho_{3n} \\ \ldots & & & & \\ \rho_{n1} & \rho_{n2} & \rho_{n3} & \ldots & 1 \end{bmatrix}$$

$\rho = \mathrm{Corr}(X)$

The scale of the data doesn't matter → [bound from -1 to 1]

Steps 3 & 4 Modeling & Analysis

Predictive Modeling → $Y = f(X)$

- Y is the outcome → can be quantitative (value) or classification (likelihood of the outcome or actual prediction)
- X is the set of predictors
- $f(X)$ is the function that maps the predictors to the outcomes
- Models can be parametric or non-parametric

Parametric Model:
$f(X)$ is restricted by parameters and relies on assumptions about these parameters (e.g., $f(X)$ is linear, Y is normally distributed, e.g., linear or logistic regression) → need to test if assumptions hold in a model

Non-Parametric Model:
$f(X)$ is not restricted by parameters → no need to test assumptions (e.g., KNN., trees, neural networks)

Desirable Predictive Model Properties:

- Interpretable → results explain relationships between Y and X well
- Accurate → predictions of Y are very close to actual values of Y
- Unbiased → the effects of X on Y are close to the true effects
- Low Variance → different data samples yield similar effects of X on Y

Analytics Modeling Options

	Modeling Method		
	Structured		Unstructured: Visual, Text, etc.
Descriptive	Cluster analysis, correlation, market basket analysis, sample statistics, ANOVA		Bubble charts, network diagrams, natural language processing, clustering dendrograms, etc.
Predictive	Association	Decision Tree	Charts
Quantitative Value	Regression	Regression Trees	Regression plots, scatter plots, Tableau diagrams, trend charts, etc.
Classification	Logistic Regression; Other Categorical Regression Models	Classification Trees	Tree maps, interactive diagrams,
Prescriptive	Operations research, decision modeling, optimization, linear programming		Simulations, etc.

s Modeling Options

- Ordinary Least Squares
- Weighted Least Squares
- Generalized Linear Model
- Ridge Regression
- LASSO Regression
- Principal Components Regression
- Partial Least Squares
- Non-Linear Models
- Piecewise Linear Regression
- Step Regression
- Linear, Cubic, Smoothing Splines
- Neural Networks
- Structural Equation Modeling
- Etc.

Modeling Method

Structured

sis, correlation,
sample statistic

- Regression Trees
- Classification Trees
- Bootstrap Aggregation (Bagging)
- Random Forest
- Boosting Trees
- Ensemble Models
- Etc.

- Binomial Logistic Regression
- Multinomial Logistic Regression
- Probit Regression
- Linear Discriminant Analysis
- Quadratic Discriminant Analysis
- Neural Networks
- Etc.

	on		
Quantitative Value	Regression		catter rams, c.
Classification	Logistic Regression; Other Categorical Regression Models	(	ctive
Prescriptive	Operations research, decision modeling, optimization, linear programming		Simulations, etc.

Predictive Modeling Goals

- Predictive modeling is not just about making predictions
- It all depends on what your analysis goals are
- Understanding the analytics question(s) you are trying to answer will give you clarity about the analysis goals
- Defining the specific goal(s) of your analysis will help you decide which modeling method is most appropriate.
- Your analysis goal could be one or more of these 3:

1. Interpretation
2. Inference
3. Prediction

Modeling Goal 1: Interpretation

- Interpretation is about explaining what the model results are telling us
- For example:
  - Holding weight, size and cylinder size constant (i.e., controlling for), adding one more vehicle cylinder reduces gas mileage by 2 mpg
  - Holding weight, exercise activity and cholesterol constant (i.e., controlling for), smoking one additional cigarette per day increases the chances of a heart attack by 0.5%
- Some methods are great for interpretation (e.g., OLS regression, logistic regression); others are not (e.g., regression trees, support vector machines)

Modeling Goal 2: Inference

- Inference is about hypothesizing an effect and testing it to see if the data supports it, for example:
  - Increased advertising leads to higher sales
  - Increased minimum wage leads to less unemployment
  - More years of college education leads to higher income
  - Low amounts of aspirin reduces heart disease

- Testing hypotheses → null and alternative hypotheses:
  - H_0: The null hypothesis (i.e., there is no effect)
  - H_0 cannot be accepted → either reject or fail to reject it
  - H_A: Alternative hypothesis is supported if H_0 is rejected
  - It is better to set H_A as what we are trying to prove, e.g.,
    - ✓ H_0: $\beta = 0$ → no effect
    - ✓ If rejected, then H_A: $\beta \neq 0$ → effect is significant

Modeling Goal 3: Prediction

- Which model is most accurate at predicting outcomes? e.g.,
  - What are the chances that a credit card charge is fraudulent?
  - What are the chances that a new email is spam?
  - How much income is a college graduate from Kogod expected to earn 5 years after graduation?
- Machine learning helps evaluate predictive accuracy:
  - Train the model using part of the data, then
  - Test the resulting model on the rest of the data
  - Cross-validation: test the accuracy (or error) of the trained model using the test data
  - Re-sample, re-train and re-test as needed
  - Select the model with the lowest test error or deviance

Model Explainability

Explainable Models

Make it easy to explain how the model works and what the results mean → e.g., OLS regression, logistic regression, decision trees → good for interpretability and inference

Black Box Models

The complexities of the internal model are difficult to explain → e.g., neural networks, deep learning → good for predictive accuracy

Modeling Method and Specification

Modeling Method:

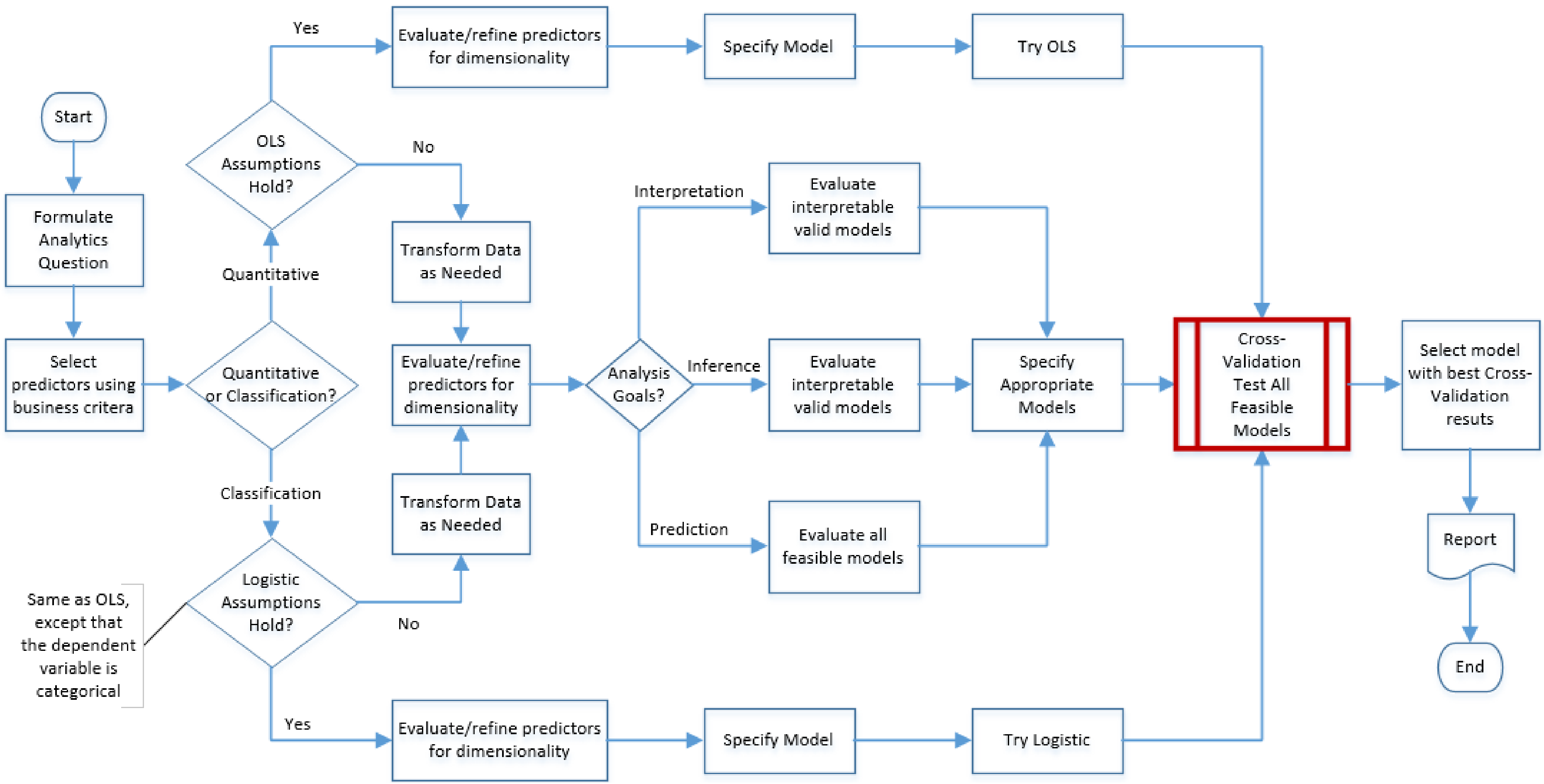
Is the particular modeling approach used to analyze the data – e.g., random forest tree, logistic regression, linear regression, Ridge regression, neural network, etc.

Model Specification:

For any given modeling method, there are many choices on how to specify a model (feature engineering). The specification of a model involves:

- Variable selection → selecting which outcome variables and predictors to use in the model
- Variable transformations → whether and how to transform specific variables either to meet parametric model assumptions or to improve the fit and interpretability of the model (e.g., polynomial, logs, etc.

Analytics Model Selection Process



Step 5 Interpretation & Storytelling

What yields the best predictive model?

- The most sophisticated model?
- The most complex specification?
- The most parsimonious (i.e., simple) model/specification?
- The least biased model?
- Model with least variance?
- The model that better fits the data?
- The most accurate model?
- The most interpretable model?
- The model that fulfills the data and model assumptions?
- The highest data quality possible?
- The most appropriate (business) selection of predictors?
- Quadratic and other transformations that fit data the best?
- All of the above?

→ It depends on your analytic goals !!

Interpretation & Storytelling

- Data doesn't convey emotion, storytelling does
- If the goal is predictive accuracy, then interpretation and storytelling are less important → machine learning and cross-validation testing are more important → e.g., read handwritten zip codes, train self-driving vehicles, tumor detection, etc.
- In most business predictive models, the goal is to interpret results for managers and clients to aid decision making
  - ✓ Evaluate the model as a whole:
    - ➢ Fit statistics, R^2, model error, error rates, deviance, etc.
    - ➢ Bias and variance of the model
  - ✓ Evaluate individual predictors
    - ✓ Effect → i.e., how much each predictor affects the outcome
    - ✓ Significance → i.e., probability of being correct
    - ✓ Variable importance → for non-statistical models

Final Notes

Final Notes

- The main goal in this course is for you to learn how to formulate a question, then select, justify and employ the most appropriate data, modeling method and specification to answer the predictive analytics question, and interpret the results in a way that makes sense to managers
- We will cover some methods in depth
- And provide only a high-level overview of others
- You will also learn how to use R for modeling methods covered in the course, but you will also be able to explore other modeling methods in R or Python on your own

KEY: Become a good analytics learner !! → we can't possibly cover the abundance of methods available → use the foundational concepts you will learn in this class to learn many other modeling approaches on your own